Agile Flow - Final Team Presentation

A Productivity Visualization Agile Methodologies Tracker

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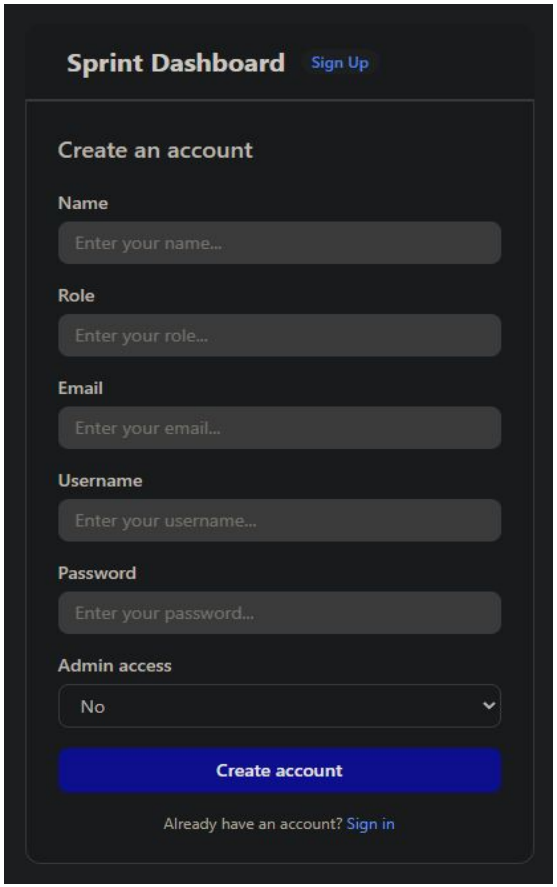
Course: CSPB 3308 Software Development Tools & Methods

Problem Motivation

- Students struggle to develop Agile Methodologies:
  - Pay-walling from current Agile productivity applications
  - Limited access to Agile implementation
  - No shared task tracking/limited resources to do so
  - Students struggle to effectively apply agile methodologies
- Goals
  - Reduce friction in team collaboration
  - Streamline Agile-based project coordination

What Agile Flow Does

- Agile Flow enables groups to:
  - Create and join projects
  - Maintain shared task boards
  - Creating projects, sprints and tasks
  - Burn down/up Chart visualize
    - scope creep
    - Track Sprint Progress
    - completion dates

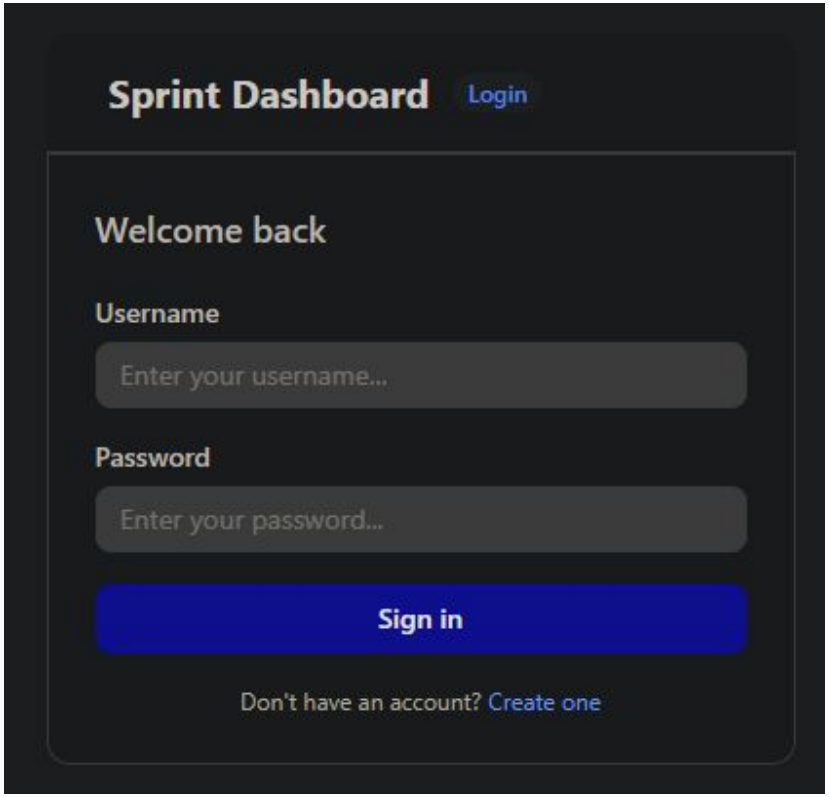


Main dashboard showing active groups and tasks.

Demo Flow Overview

- Our live demo will follow this sequence:

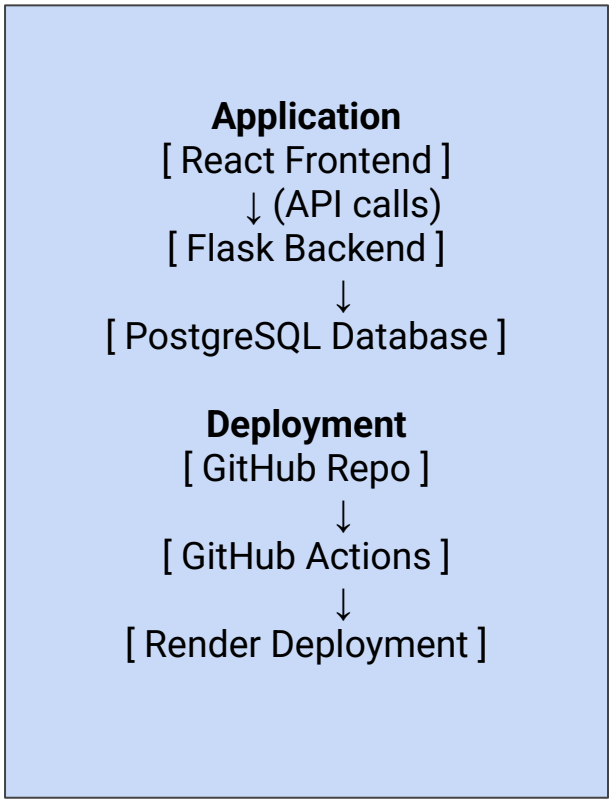
1. Display Login Page
2. Display Create Account Page
3. Fetch current user data (mock login)
4. Dashboard Overview
    a. Create a Project
    b. Create a Sprint
    c. Create Multiple Tasks
    d. Update Task Completeness
    e. Delete Tasks
    f. Delete a Sprint



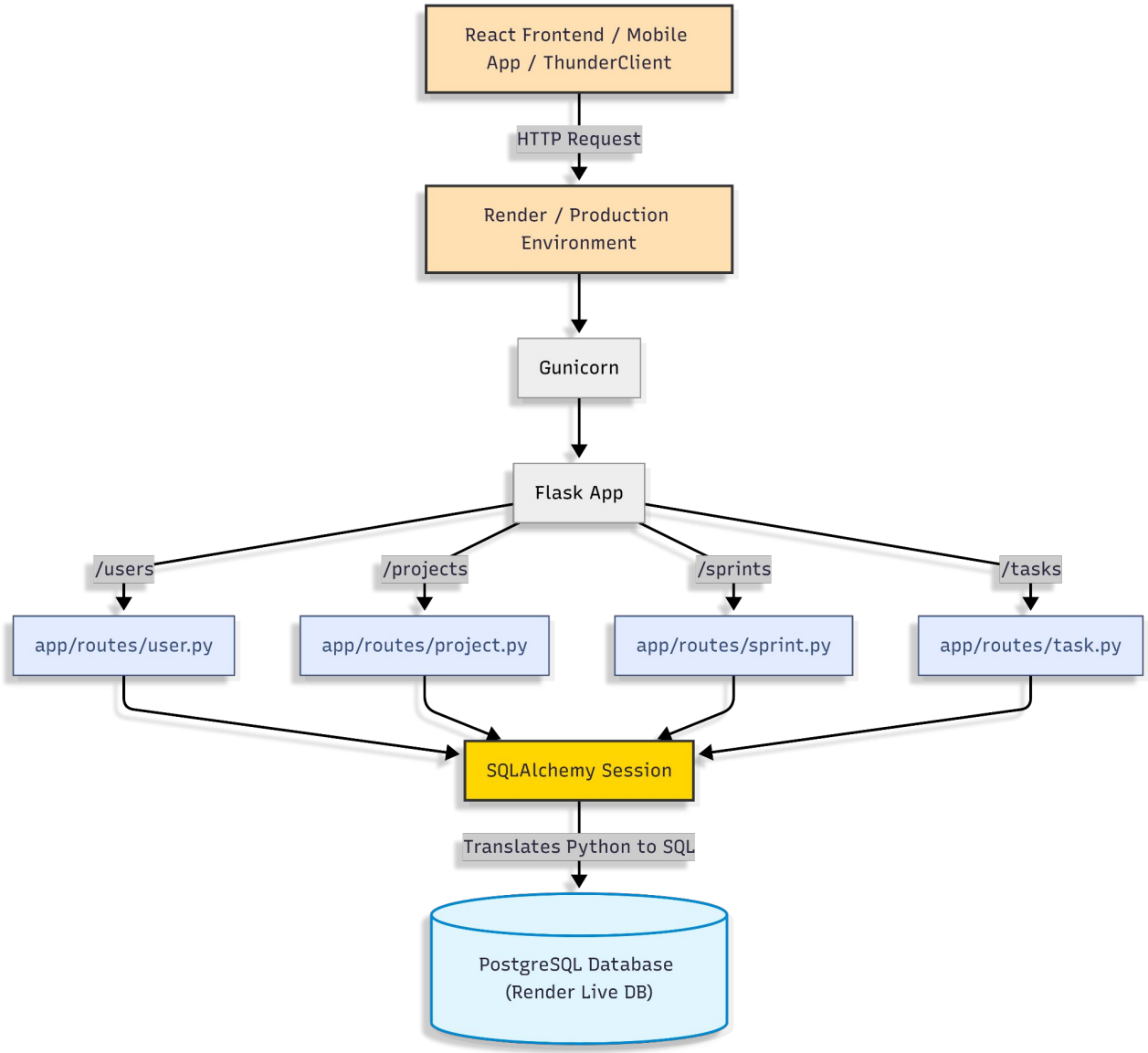
Login screen with authentication flow.

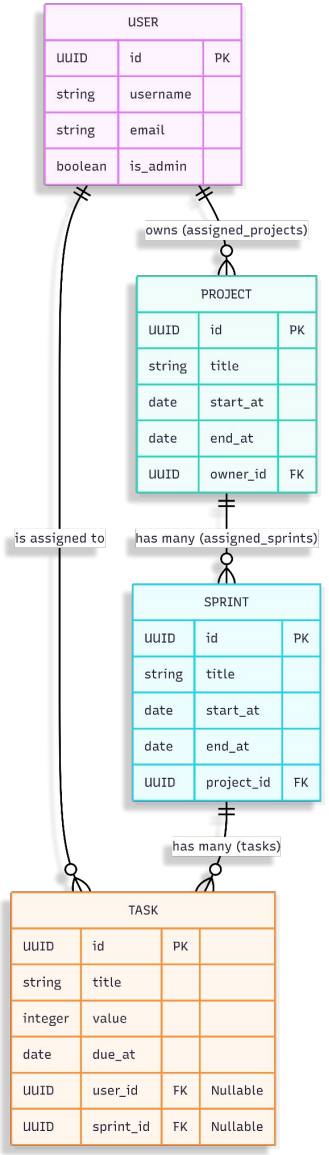
System Architecture Overview

- FRONTEND:
  - React.JS w/Tailwind CSS
  - ChartsJS
- BACKEND:
  - Flask RESTful API
- DATABASE:
  - PostgreSQL - SqlAlchemy
  - ORM
- CI/CD:
  - GitHub Actions → Render

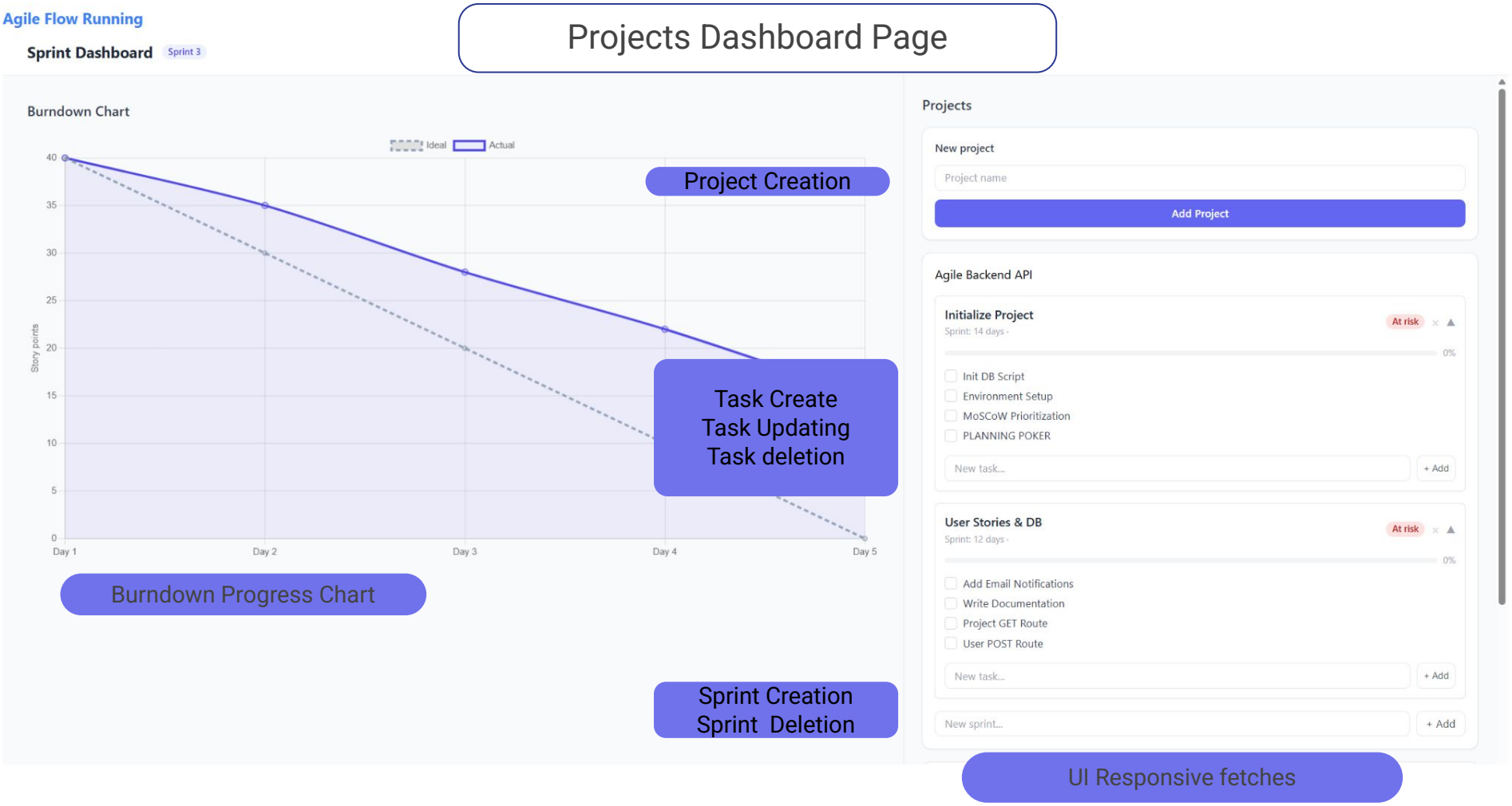


Architecture Diagram



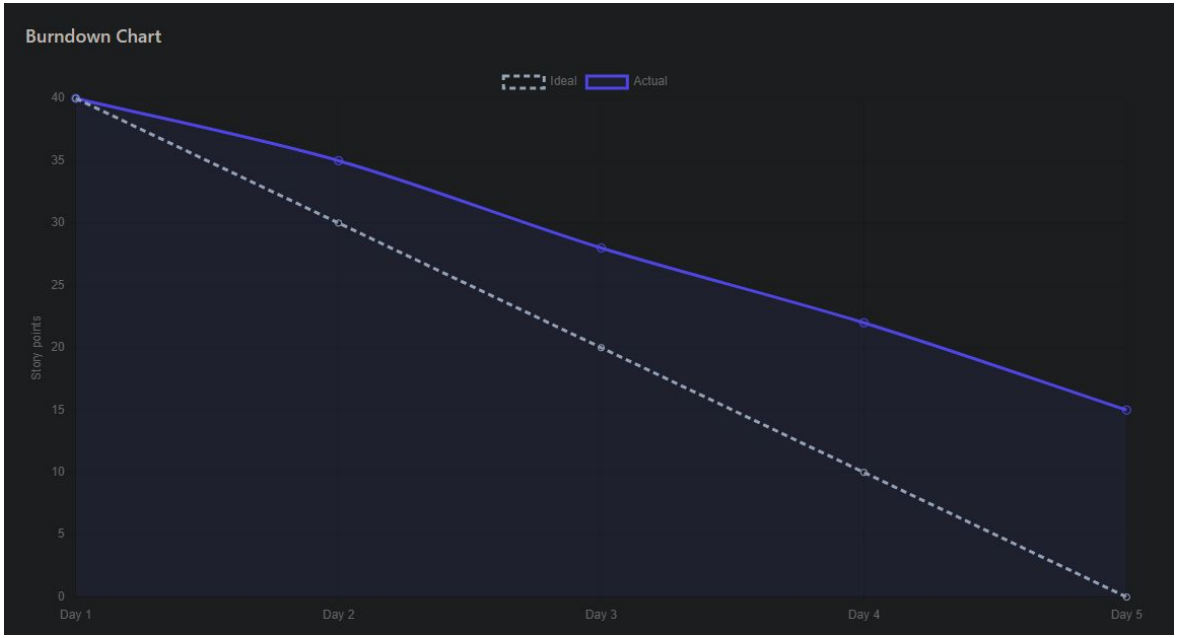


Key Features Implemented



Burndown Chart Feature

- Visualizes sprint progress over time
- Compares planned vs completed work
- Helps team track velocity and workload (tasks/sprints)
- Updates when tasks are completed

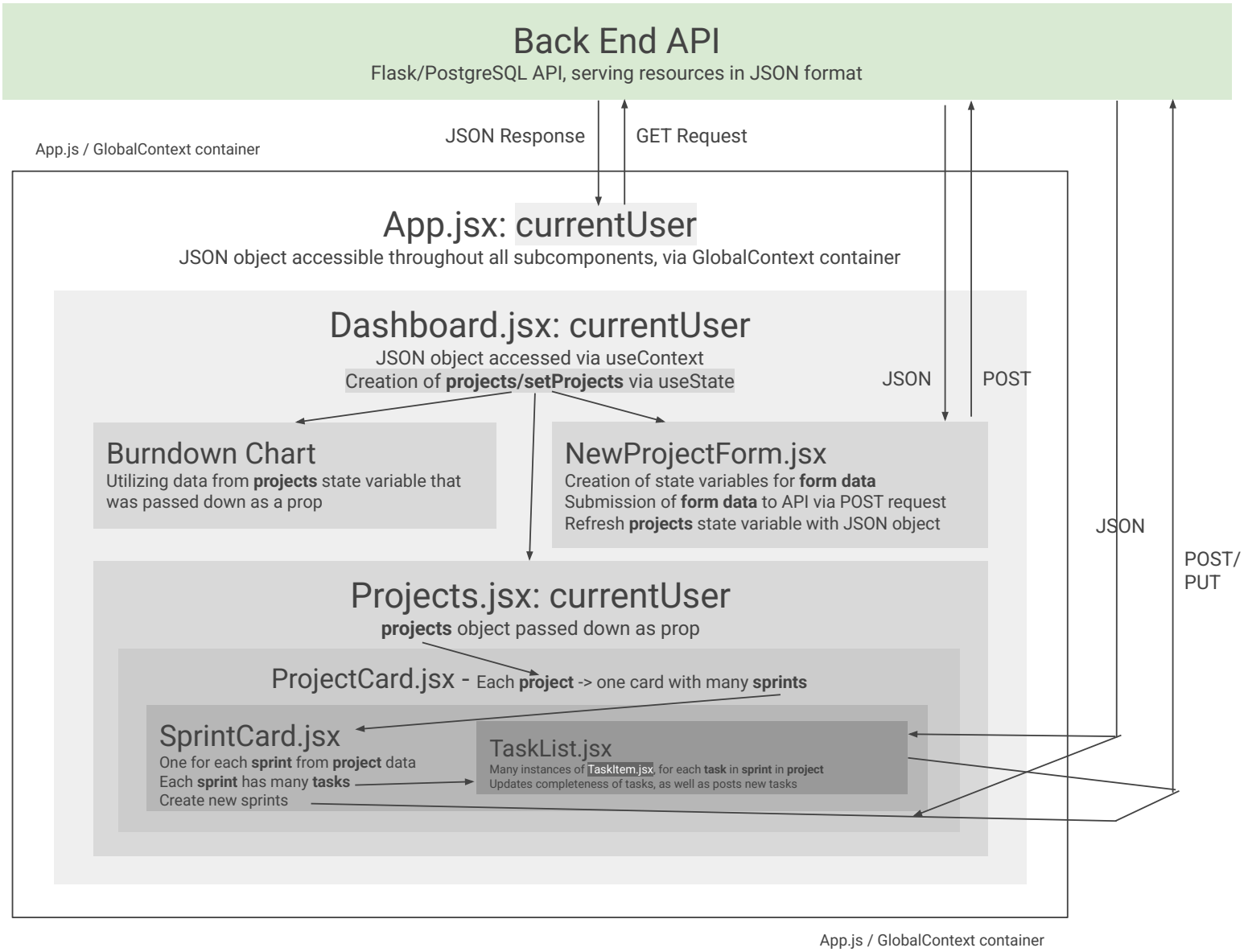


Burndown Chart showing sprint progress over time

Major Technical Challenges

1. Headless API Deployment
    a. Creates hurdles in information flow/interpretation between different systems
2. State management in React *(diagram on next slide)*
    a. Component trees and resultant API fetches can create challenges in info management
3. Render Deployment
    a. Slow spin up time can postpone testing by a few minutes
    b. Deployment sometimes crashes on a imited free resource such as Render
4. Local PostGres Database (pgAdmin)
    a. Trying to create a local database is a convoluted process
5. CI/CD workflow debugging

Diagram: Front End State Variable Flow of Information



What Went Well

- Team communication using Discord, weekly meetings, and Jira for task tracking
- Effective sprint-based collaboration with regular weekly updates and reviews
- Continuous integration of work through coordinated frontend/backend development
- Informal usability testing across components improved overall system design
- Rotating Scrum Master role helped distribute leadership and coordination

What Didn’t Go Well

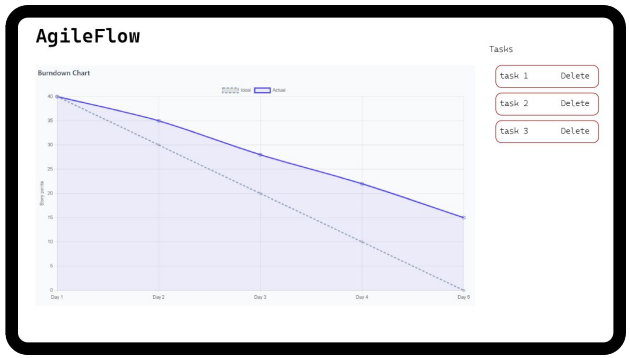
- Frontend and backend integration took longer than expected
- Delays in initial sprint planning while team roles were being finalized
- Some Learning curves with different programming languages
- Some tasks required rework during integration and testing phases
- Overall, we did not accomplish all our goals in terms of building out components
  - Burndown chart is not connected to the data
  - Account creation/login authentication with user sessions is not built out

Lessons Learned

- Using Jira helped improve sprint organization and task tracking
- GitHub branching and pull requests are critical for team-based development
- Clear role division (frontend/backend) improves development efficiency
- Clear communication prevents technical drift
- Agile sprint structure improves adaptability and project flow

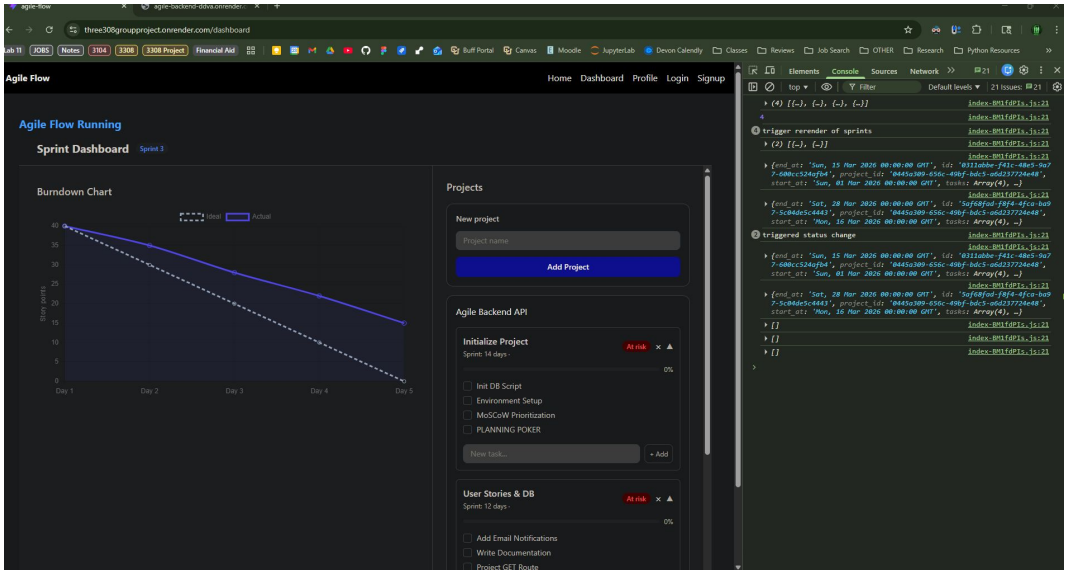
Future Work

- If we continued development, we would add:
  - Messaging in between users
  - Push notifications
  - Calendar integrations
  - Rich messaging features
  - Detailed user profiles



Final Demo Highlights

- We will highlight:
  - Demonstration of successful fetches, logging various data objects in the console
  - Smooth navigation of site, with sleek Tailwind CSS UI elements
  - Dynamic population of page information using fetched data
  - Effective CRUD operations on multiple data models, with persistence after refresh



Final Reflection

- We learned how to:
  - Plan and deliver iterative milestones
  - Collaborate using GitHub effectively
  - Evaluate architectural tradeoffs
  - Prepare a polished full-stack project
  - Present technical work professionally

Thank You

- Questions?
- GitHub:
  - Front End: https://github.com/Agile9-3308/3308GroupProject
  - Back End: https://github.com/Sergrojas29/Agile-Backend
  - Deployed Front End: https://three308groupproject.onrender.com/
  - Deployed Back End: https://agile-backend-ddva.onrender.com/